

Real analysis Week 1 Lecture 1 and 2 homework

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1 Problem 3.6

1.1 a) Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Let $x \in A \cap (B \cup C)$ therefore, $x \in A$ and $x \in B \cup C \Rightarrow x \in A \cap B$ or $x \in A \cap C \Rightarrow x \in (A \cap B) \cup (A \cap C)$ therefore, $A \cap (B \cup C) \subset (A \cap B) \cup (A \cap C)$

Let $x \in (A \cap B) \cup (A \cap C)$ therefore, $x \in A \cap B$ or $x \in A \cap C \Rightarrow x \in A$ and $x \in B \cup C \Rightarrow x \in A \cap (B \cup C)$ Therefore, $(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C)$ therefore,

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad \square$$

1.2 b) Prove $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Let $x \in A \cup (B \cap C)$ therefore, $x \in A$ or $x \in B \cap C \Rightarrow x \in A \cup B$ and $x \in A \cup C \Rightarrow x \in (A \cup B) \cap (A \cup C)$ therefore, $A \cup (B \cap C) \subset (A \cup B) \cap (A \cup C)$

Let $x \in (A \cup B) \cap (A \cup C)$ therefore, $x \in A \cup B$ and $x \in A \cup C \Rightarrow x \in A$ or $x \in B \cap C \Rightarrow x \in A \cup (B \cap C)$ Therefore, $(A \cup B) \cap (A \cup C) \subset A \cup (B \cap C)$ therefore,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \quad \square$$

2 Problem 3.11 Prove $n < 2^n$ by induction

Base case ($n=1$): $1 < 2^1$

Induction step: Given $n \in \mathbb{N}$ assume $2^n > n$ since $2^{n+1} = 2^n \cdot 2$, $2^{n+1} > 2n = n+n$. If $n > 1$, then $n+n > n+1$, and therefore $2^{n+1} > n+1$.

By induction for $n \in \mathbb{N}$

$$2^n > n \quad \square$$

3 Problem 3.11 Show that for a finite set A of cardinality n , the cardinality of $\mathcal{P}(A)$ is 2^n

Let $A = \{x_1, x_2, \dots, x_n\}$ have n elements, $\forall y \in \mathcal{P}(A), \exists x \in A$, there are 2 possibilities, either x will be or will not be in y . Since A has n elements, $\mathcal{P}(A)$ has $\overbrace{2 \cdot 2 \cdots 2}^n = 2^n$ elements. Therefore $|\mathcal{P}(A)| = 2^n \quad \square$.

4 Problem 3.15 Prove that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

We are going to prove this by induction.

Base case: for $n=1$ $1^3 + 5 = 6$ and $6|6$.

Induction step: Assume $6 | n^3 + 5n$ therefore $(n+1)^3 + 5(n+1) = n^3 + 3n^2 + 8n + 6 = n^3 + 5n + 3n^2 + 3n + 6$, since, $6 | n^3 + 5n$ and $6 | 3n^2 + 3n + 6 \equiv 3n^2 + 3n + 6 \pmod{6}$

Either n is even, or n is odd. If n is even, n can be written as $n = 2m$, where $m \in \mathbb{N}$. Therefore $3n^2 + 3n + 6 = 12m^2 + 6m + 6$ and $6 \mid 12m^2 + 6m + 6$. If n is odd, $n = 2m + 1$ then $3n^2 + 3n + 6 = 3(2m + 1)^2 + 3(m + 1) + 1 = 12m^2 + 18m + 12$ and $6 \mid 12m^2 + 18m + 12$. Therefore $6 \mid (n + 1)^3 + 5(n + 1)$ if $6 \mid n^3 + 5n$.

By induction $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$. $\square$

5 Problem 3.19 Give an example of a countable collection of finite sets $A_1, A_2, \dots$, whose union is not a finite set

Let the sequence of $A_1, A_2, \dots$, be defined by $A_n = \{n : 2^{n-1} < n \leq 2^n\}$. The union of the sets $A_1 \cup A_2 \cup \dots = \mathbb{N}$. While the cardinality of the set of the sequence $|(A_1, A_2, \dots)| = |\mathbb{N}|$ and $\mathbb{N}$ is countable. $\square$

6 Prove $|\{q \in \mathbb{Q} : q > 0\}| = |\mathbb{N}|$

Let there be a function $f : \{q \in \mathbb{Q} : q > 0\} \rightarrow \mathbb{N}$, $f(1) = 1$. If $q \in \mathbb{N} \setminus \{1\}$

$$f(q) = p_1^{2r_1} p_2^{2r_2} \dots p_N^{2r_N}$$

and if $q \in \mathbb{Q} \setminus \mathbb{N}$, then

$$f(q) = p_1^{2r_1} p_2^{2r_2} \dots p_N^{2r_N} q_1^{2s_1-1} q_2^{2s_2-1} \dots q_N^{2s_N-1}$$

Let $x_1, x_2 \in \mathbb{Q}$ where $f(x_1) = f(x_2)$, if $x_1 = x_2 = 1$ then $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Else if $x_1, x_2 \in \mathbb{N} \setminus \{1\}$, Let $x_1 = p_{a1}^{r_{a1}} p_{a2}^{r_{a2}} \dots p_{aN}^{r_{aN}}$ and $x_2 = p_{b1}^{r_{b1}} p_{b2}^{r_{b2}} \dots p_{bN}^{r_{bN}}$, since $f(x_1) = f(x_2)$

$$p_{a1}^{2r_{a1}} p_{a2}^{2r_{a2}} \dots p_{aN}^{2r_{aN}} = p_{b1}^{2r_{b1}} p_{b2}^{2r_{b2}} \dots p_{bN}^{2r_{bN}}$$

$$(x_1)^2 = (x_2)^2 \Rightarrow x_1 = x_2$$

Finally if $x_1, x_2 \in \mathbb{Q} \setminus \mathbb{N}$, let $x_1 = \frac{p_{a1}^{r_{a1}} p_{a2}^{r_{a2}} \dots p_{aN}^{r_{aN}}}{q_{a1}^{s_{a1}} q_{a2}^{s_{a2}} \dots q_{aN}^{s_{aN}}}$ and $x_2 = \frac{p_{b1}^{r_{b1}} p_{b2}^{r_{b2}} \dots p_{bN}^{r_{bN}}}{q_{b1}^{s_{b1}} q_{b2}^{s_{b2}} \dots q_{bN}^{s_{bN}}}$, since $f(x_1) = f(x_2)$

$$p_{a1}^{2r_{a1}} p_{a2}^{2r_{a2}} \dots p_{aN}^{2r_{aN}} q_{a1}^{2s_{a1}-1} q_{a2}^{2s_{a2}-1} \dots q_{aN}^{2s_{aN}-1} = p_{b1}^{2r_{b1}} p_{b2}^{2r_{b2}} \dots p_{bN}^{2r_{bN}} q_{b1}^{2s_{b1}-1} q_{b2}^{2s_{b2}-1} \dots q_{bN}^{2s_{bN}-1}$$

$$(x_1)^2 q_{a1}^{s_{a1}} q_{a2}^{s_{a2}} \dots q_{aN}^{s_{aN}} = (x_2)^2 q_{b1}^{s_{b1}} q_{b2}^{s_{b2}} \dots q_{bN}^{s_{bN}}$$

Since all the numbers in this factorization are prime $q_{a1}^{s_{a1}} q_{a2}^{s_{a2}} \dots q_{aN}^{s_{aN}} = q_{b1}^{s_{b1}} q_{b2}^{s_{b2}} \dots q_{bN}^{s_{bN}}$. Therefore $x_1 = x_2$ and the function is injective.

Want to show the function is surjective, $\forall y \in \mathbb{N}$ there exists a unique prime factorization $y = p_1^{r_1} p_2^{r_2} \dots p_N^{r_N}$. For r_n let $r_n = 2a_n$ if r_n is even, else let $r_n = 2b_n - 1$, therefore y can be written as

$$y = p_1^{2a_1} p_2^{2a_2} \dots p_n^{2a_n} p_{n+1}^{2b_1-1} p_{n+2}^{2b_2-1} \dots p_N^{2b_n-1}$$

Since for any y , there exists an x such that $f(x)=y$, where $x = \frac{p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}}{q_{n+1}^{b_1} q_{n+2}^{b_2} \dots q_N^{b_n}}$. Since f is both surjective and injective, f is bijective and therefore

$$|\{q \in \mathbb{Q} : q > 0\}| = |\mathbb{N}| \quad \square$$